4.1.1 Reference data

Reference data represents relatively static information about physical assets, organizational structures, and system configurations. This data provides context for interpreting operational measurements and includes asset hierarchies, device specifications, maintenance procedures, organizational charts, and configuration parameters.

Unlike operational data that changes continuously, reference data typically remains stable for weeks, months, or years. A pump's manufacturer, model number, and installation date rarely change, while its operational parameters like flow rate and temperature vary continuously. This stability influences both storage requirements and update frequencies.

Reference data often originates from enterprise systems like ERP platforms, asset management databases, and configuration management tools. The master data management challenge lies in maintaining consistency across multiple systems that may store overlapping information about the same physical assets.

4.1.2 Timeseries data

Timeseries data represents the continuous flow of measurements, observations, and state changes that characterize system behavior over time. This operational data is what allows a digital twin to keep its digital representation of the physical system updated and accurate, enabling monitoring, trend analysis, and predictive modeling that drives automated decision-making.

Timeseries data exhibits several key characteristics:

- *Temporal ordering*—Each measurement is associated with a precise timestamp that establishes when the observation occurred, enabling analysis of trends, sequences, and causality relationships.
- *Temporal locality*—Recent data typically requires more frequent access than historical data, influencing storage and indexing strategies.
- *High frequency and volume*—Sensors produce continuous streams ranging from thousands of measurements per second in industrial equipment to periodic readings every few minutes in smart home applications. The continuous generation of data by many sensors creates large volumes of data that require specialized storage and processing approaches optimized for temporal queries and aggregation operations.
- *Immutable*—Historical measurements represent facts that rarely change once recorded, unlike transactional data that frequently updates.

Table 4.1 shows an example of timeseries data produced by a single sensor in my home that measures a range of environmental parameters, showing the characteristics that are typical of this type of data.